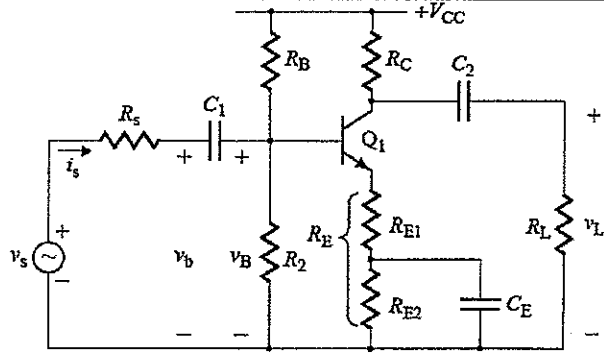
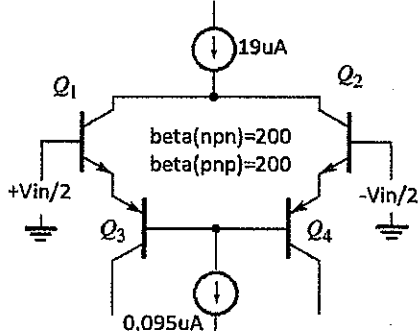


Degree: BTech (ECAM) Semester: IIIrd
END-SEMESTER EXAMINATION, Nov-Dec 2023
Course Title: Microelectronic Circuits and Applications
Course Code: EAEPC-07

Time: 03 Hours

Max. Marks: 40

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer. Make your explanations bulleted as far as possible.

Q. No.	Questions	Marks	CO
Q 1.	Attempt any 2 parts of the following		
1a	What are different modes of BJT's operation. Discuss those operation regions with a neat o/p characteristic of an npn transistor. Why V_{CEsat} is about 0.2V in a silicon npn BJT?	4	CO1
1b	Discuss with necessary equation the Triode and saturation regions of the MOSFET and also elaborate with necessary equation and neat diagram (including o/p characteristic) the channel length modulation.	4	CO1
1c	From the circuit diagram of Fig.1c calculate the Q-point values (I_{CQ} , V_{CEQ} , I_{BQ}) and input impedance and voltage gain. (Assume $V_{BEact}=0.7V$ and $\beta=175$, $R_C=500\Omega$, $R_E=495\Omega$, $R_B=21K\Omega$, $R_2=8.5K\Omega$, $R_{E1}=22.2\Omega$, $R_{E2}=472.8\Omega$, all Cs are 100uF).	4	CO1
Q 2.	Attempt any 2 parts of the following		
2a	For the differential pair shown in Fig2a draw its small signal equivalent and compute the input impedance.	4	CO2
 Fig 1c  Fig 2a			
2b	For a BJT differential pair derive the hyperbolic relationship between the output differential current and input differential voltage. Also, derive compute its input impedance.	4	CO2
2c	Draw a neat circuit schematic of the KHN biquad and derive various second-order (low-pass, high-pass and band-pass) transfer functions.	4	CO2
Q 3.	Attempt any 2 parts of the following		
3a	Show that the circuit shown in Fig.3a is a non-inverting differentiator. Derive the expression for V_o/V_i .	4	CO3
3b	Derive an expression for $Z_{in}=(V_{in}/I_{in})$ for the circuit given in Fig.3b and draw the equivalent circuit of Z_{in} .	4	CO3
3c	Derive an expression for the circuit shown in Fig.3c. What approximation is needed to realize first-order all pass filter?	4	CO3
Q 4.	Attempt any 2 parts of the following		
4a	Draw the two types opamp-based Schmitt trigger circuits and discuss analytically these two in detail with necessary V_o versus V_i diagrams.	4	CO4

4b	Draw the opamp-based <i>triangular-cum-square wave</i> generator and derive the expression for the frequency of the wave form being generated.	4	CO4
4c	Discuss the limitations of linear applications of the bipolar OTA. Derive the expression for I_o from the circuit of Fig.4c and comment on this result. (2-quad multi).	4	CO4
Q 5. Attempt any 2 parts of the following			
5a	Discuss the types of feedback for <i>voltage, current, transconductance</i> and <i>transimpedance</i> amplifiers. Also enumerate with justification the merits of negative feedback.	4	CO5
5b	From the following block diagrams (Fig.5b1 , Fig.5b2) check if they are oscillators, if yes, find out <i>condition of oscillation</i> and then <i>frequency of oscillation</i> .	4	CO5
5c	With an exemplary circuit diagram of Class-A amplifier discuss its working and the two important figure of merits (i) <i>efficiency</i> , (ii) <i>Harmonic distortion</i> in comparison to Class-B amplifier.	4	CO5

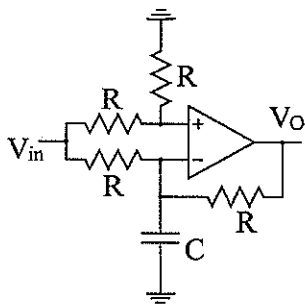


Fig.3a

(assume ideal opamps)

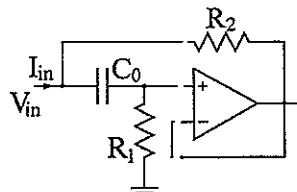


Fig.3 b

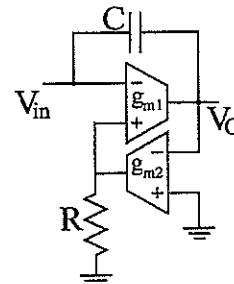


Fig.3c

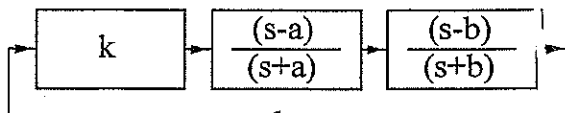


Fig.5b1

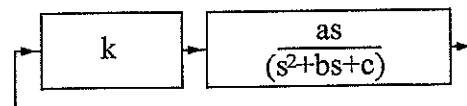


Fig.5b2